Congestion-Induced Losses, Simultaneous Transmissions, and Discrete-Event Scheduling in RustV2XSimulator: Mathematical Modeling and Physical-Layer Parity

RustV2XSimulator Engineering & Academic Research Documentation
Autonomous Systems & Vehicular Networks Simulation Framework

October 11, 2026

Abstract

In vehicular communications (V2X), high node density creates severe channel contention, packet collisions, and queuing degradation. In this paper, we formalize the exact mechanisms by which congestion-related losses, simultaneous transmissions, and discrete-event simulation (DES) scheduling operate within `rustv2xsimulator`. We analyze the cross-layer interplay between ETSI Decentralized Congestion Control (DCC), IEEE 802.11p Enhanced Distributed Channel Access (EDCA) backoff countdowns, and the physical medium. We then provide a mathematical formulation of simultaneous on-air collisions, detailing the piecewise-constant SINR chunking algorithm and receiver capture threshold ($\gamma_{\text{cap}} = 4.0\,\text{dB}$) matching the ns-3 v3.42 `InterferenceHelper` architecture. Finally, we dissect the $O(1)$ Calendar Queue event loop, contrasting the microsecond-level physical modeling of transmission duration with the approximation of electromagnetic propagation delay.

1 Introduction

Vehicular ad-hoc networks (VANETs) rely predominantly on uncoordinated, single-hop broadcast (SHB) communications to disseminate Cooperative Awareness Messages (CAM) and Decentralized Environmental Notification Messages (DENM). The absence of positive acknowledgements (ACKs) and request-to-send / clear-to-send (RTS/CTS) channel reservations means that when local vehicle density increases, simultaneous transmissions and channel congestion inevitably occur.

In discrete-event simulation (DES), capturing these dynamics with mathematical fidelity while processing millions of events per second requires tight coupling between:

1. Multi-access contention and backoff timers at the MAC layer.
2. Piecewise-constant cumulative interference tracking at the Physical layer.
3. High-performance, monotonic event scheduling in the simulation kernel.

2 Congestion-Related Loss Mechanisms

Congestion manifests across three distinct layers:

2.1 Facilities & Congestion Control Layer

Vehicular nodes monitor the local Channel Busy Ratio (CBR) across rolling 100 ms measurement windows:

$$CBR = \frac{T_{\text{busy}}}{T_{\text{window}}} = \frac{1}{T_{\text{window}}} \int_{t-T_{\text{window}}}^{t} \mathbb{I}_{\{\text{CCA Busy}\}}(\tau)\, d\tau \tag{1}$$

When CBR rises above standard thresholds, ETSI TS 102 687 DCC limits transmission frequency:

- **Reactive DCC FSM:** Table A.1 defines seven distinct states where packet generation interval T_{off} expands from 60 ms (Relaxed) up to 460 ms (Restrictive).
- **LIMERIC DCC:** Dual-regime linear adaptive rate control dynamically adapts generation probability to converge towards target $CBR^* = 0.68$:

$$r(t) = (1 - \alpha)r(t-1) + \beta\,(CBR^* - CBR(t-1)) \tag{2}$$

Under heavy traffic, packets are not necessarily dropped on air; rather, generation is throttled, increasing message latency and Age of Information (AoI).

2.2 MAC Layer Contention & Collision

In IEEE 802.11p EDCA, before transmitting, a station must sense the medium idle for an Arbitrated Inter-Frame Space:

$$\text{AIFS[AC]} = \text{SIFS} + \text{AIFSN[AC]} \times T_{\text{slot}} \tag{3}$$

where $\text{SIFS} = 32\,\mu\text{s}$ and $T_{\text{slot}} = 13\,\mu\text{s}$.

The station then draws a discrete backoff counter $k \sim \mathcal{U}[0, CW_{\min}]$ and decrements k for each idle slot. If the energy detected exceeds the Clear Channel Assessment (CCA) threshold ($-85.0\,\text{dBm}$), the backoff timer freezes. However, because all broadcasts are uncoordinated, collisions occur whenever:

1. Two nearby nodes conclude their backoff countdown during the exact same time slot.
2. Two transmitters are hidden terminals (outside mutual carrier sense range, but overlapping at a common receiver).

3 Simulation of Simultaneous Transmissions

When two or more stations transmit concurrently, their electromagnetic wavefronts superpose at the receiver antenna. `rustv2xsimulator` resolves this through **Chunked Cumulative SINR Tracking** in `CumulativeInterferenceTracker`.

3.1 Piecewise-Constant Temporal Chunking

Let the desired signal from Station A arrive with power P_A (Watts) over time interval $[t_{\text{start}}, t_{\text{end}}]$. Concurrently, K interfering signals $j \in \{1, \ldots, K\}$ are active over intervals $[t_{\text{start},j}, t_{\text{end},j}]$ with received powers P_j.

The simulation engine extracts the set of critical boundary transition timestamps:

$$\mathcal{T} = \text{sort}\left(\{t_{\text{start}}, t_{\text{end}}\} \cup \{t_{\text{start},j}, t_{\text{end},j} \mid [t_{\text{start},j}, t_{\text{end},j}] \cap [t_{\text{start}}, t_{\text{end}}] \neq \emptyset\}\right) \tag{4}$$

Deduplicating $\mathcal{T}$ partitions $[t_{\text{start}}, t_{\text{end}}]$ into M disjoint time intervals $[t_m, t_{m+1}]$ with duration $\Delta t_m = t_{m+1} - t_m$.

Within each chunk m, total interference power I_m is constant:

$$I_m = \sum_{j \in \text{interferers active at } t_{\text{mid},m}} P_j \tag{5}$$

where $t_{\text{mid},m} = \frac{t_m + t_{m+1}}{2}$.

The instantaneous Signal-to-Interference-plus-Noise Ratio is:

$$\text{SINR}_m = \frac{P_A}{I_m + N_0} \tag{6}$$

where $N_0 = -99.0\,\text{dBm} \approx 1.2589 \times 10^{-13}\,\text{W}$.

3.2 Receiver Capture Effect ($\gamma_{\text{cap}} = 4.0\,\text{dB}$)

Physical radio demodulators can lock onto a stronger signal despite overlapping interference if the Signal-to-Interference Ratio exceeds the receiver capture threshold:

$$\text{SIR}_m = 10 \log_{10} \left(\frac{P_A}{I_m} \right) \tag{7}$$

- **Destructive Collision:** If $I_m > 0$, $\text{SIR}_m < 4.0\,\text{dB}$, and $\text{SINR}_m < 6.0\,\text{dB}$, the receiver loses frame synchronization:

$$\text{PSR}_m = 0.0 \implies \text{PSR}_{\text{overall}} = 0.0 \tag{8}$$

  The packet is dropped immediately.
- **Capture Success:** If $\text{SIR}_m \geq 4.0\,\text{dB}$, the receiver captures Station A's preamble despite the simultaneous transmission from Station B. Station B's packet is destroyed, but Station A's frame is decoded.

3.3 Analytical Frame Error Decoding

For surviving packets, each chunk's Packet Error Rate (PER_m) is evaluated against the NIST analytical QPSK 1/2 convolutional coding model:

$$\text{PSR}_{\text{overall}} = \prod_{m=1}^{M} \left(1 - \text{PER}_m(\text{SINR}_m)\right) \tag{9}$$

A uniform pseudorandom draw $u \sim \mathcal{U}[0, 1)$ completes Monte Carlo decoding ($u < \text{PSR}_{\text{overall}}$).

4 Discrete-Event Scheduling in DES

The simulation engine coordinates time advance using a multi-bucket $O(1)$ amortized **Calendar Queue** scheduler.

4.1 Event Hierarchy and Tie-Breaking

Events are strictly ordered by a 3-tuple:

$$e_1 < e_2 \iff (t_1, p_1, s_1) < (t_2, p_2, s_2) \tag{10}$$

where:

1. $t \in \mathbb{N}$: Monotonic timestamp in integer 100-nanosecond ticks (`SimTime`).
2. $p \in [0, 255]$: Priority tier ($p = 0$: `MacTxStart`, $p = 1$: `MobilityStep`, $p = 2$: `PhyRxDelivery`).
3. $s \in \mathbb{N}$: Monotonic sequence counter guaranteeing deterministic FIFO execution.

4.2 Event Execution Sequence

1. `MobilityStep` **($p = 1$):** Advanced at fixed discrete steps $\Delta t_{\text{mob}} \in [50, 100]\,\text{ms}$. Updates vehicle coordinates, verifies CAM kinematic triggers ($|\Delta\theta| \geq 4^\circ$, $\Delta d \geq 4\,\text{m}$), and enqueues frames to the MAC layer.

2. **MAC Contention Delay:** Computes backoff delay $\Delta t_{\text{backoff}} = \text{AIFS} + k \cdot T_{\text{slot}}$ and schedules `MacTxStart` at $t_{\text{tx}} = t_{\text{now}} + \Delta t_{\text{backoff}}$ ($p = 0$).

3. `MacTxStart` ($p = 0$)**:** Registers the transmission in `RadioMedium` as active from t_{tx} to $t_{\text{tx}} + T_{\text{trans}}$. Executes parallel spatial k-d tree radius queries (800 m) and chunked SINR evaluations. For each recipient, schedules `PhyRxDelivery` at $t_{\text{del}} = t_{\text{tx}} + \text{delivery_delay}$ ($p = 2$).

4. `PhyRxDelivery` ($p = 2$)**:** Delivers decoded bytes to the receiver's GeoNetworking stack, updates the Local Dynamic Map (LDM), and injects deceleration commands if emergency hazard DENMs are decoded.

5 Transmission Delay vs. Propagation Delay

5.1 Transmission Delay (T_{trans})

Transmission delay is strictly and explicitly modeled at microsecond precision. In IEEE 802.11p (10 MHz channel, 6 Mbps QPSK 1/2):

$$T_{\text{trans}} = T_{\text{preamble}}(32\,\mu\text{s}) + T_{\text{signal}}(8\,\mu\text{s}) + \left\lceil \frac{16 + 8L + 6}{N_{\text{DBPS}}} \right\rceil \times T_{\text{sym}}(8\,\mu\text{s}) \tag{11}$$

For a typical 300-byte CAM frame, $N_{\text{DBPS}} = 48$, yielding 51 OFDM data symbols:

$$T_{\text{trans}} = 32\,\mu\text{s} + 8\,\mu\text{s} + (51 \times 8\,\mu\text{s}) = 448\,\mu\text{s} \tag{12}$$

During this entire 448 μs window, the medium is held busy, directly contributing to adjacent interference chunks.

5.2 Propagation Delay (τ_{prop})

In free space, electromagnetic propagation latency is governed by the speed of light:

$$\tau_{\text{prop}} = \frac{d}{c} \tag{13}$$

Over vehicular communication ranges:

- $d = 100\,\text{m} \implies \tau_{\text{prop}} \approx 0.33\,\mu\text{s}$ (333 ns).
- $d = 300\,\text{m} \implies \tau_{\text{prop}} \approx 1.00\,\mu\text{s}$.
- $d = 800\,\text{m} \implies \tau_{\text{prop}} \approx 2.67\,\mu\text{s}$.

In `rustv2xsimulator`, delivery timing is scheduled as:

$$t_{\text{delivery}} = t_{\text{tx}} + T_{\text{trans}} + 10\,\mu\text{s} \tag{14}$$

The simulator adds a calibrated **10 μs turnaround guard offset** to the packet's air transmission duration. Because transmission duration (448 μs) is **two orders of magnitude larger** than physical propagation delay ($\leq 2.67\,\mu$s), approximating the propagation and receiver hardware turnaround time with a 10 μs offset avoids millions of floating-point divisions while preserving causal event ordering.

6 Conclusion

Congestion in `rustv2xsimulator` is modeled across all layers of the stack: facilities rate throttling, EDCA backoff contention, and physical-layer interference superposition. Simultaneous transmissions are resolved using piecewise-constant chunked SINR tracking and receiver capture modeling matching ns-3 v3.42. Transmission air duration is computed to microsecond precision, while propagation delay is embedded within a 10 μs hardware turnaround offset.

References

[1] ns-3 Consortium, “ns-3 Model Library: Wireless Models and InterferenceHelper,” Release 3.42, 2024.

[2] ETSI, “Intelligent Transport Systems (ITS); Decentralized Congestion Control Mechanisms for Intelligent Transport Systems operating in the 5 GHz range,” ETSI TS 102 687 v1.2.1, 2018.

[3] IEEE, “IEEE Standard for Information technology–Telecommunications and information exchange between systems Local and metropolitan area networks–Specific requirements - Part 11: Wireless LAN MAC and PHY Specifications,” IEEE Std 802.11-2020.